D07 · Pick-Pack-Ship & Fulfillment Execution

03 · Shipping execution: carrier booking, labeling & documentation

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Shipping execution covers everything between a packed order and a departed truck: carrier booking, labels and tracking numbers, shipping documents — in Mexico, the CFDI with the Carta Porte complement — and verified loading.

2 WHY IT MATTERS

Shipping is where small errors become customer-visible failures: a wrong label sends a parcel astray, a missing document stops a truck. **In Mexico, transporting goods without a valid CFDI with Carta Porte complement exposes the shipment to retention and fines**, so document generation must come from system data, not manual typing.

3 HOW IT IS DONE

1. **Integrate WMS, TMS and billing.** Labels, carrier bookings and CFDI generated automatically from shipment data.
2. **Keep carrier master data current.** Tax ID, permits and insurance validated before a carrier is used.
3. **Verify loading by scan.** Every documented package confirmed on the truck before the dock closes.
4. **Measure documentation accuracy.** Track shipments with complete, correct documents as a 100% target.

IN PRACTICE

The CFDI de traslado with Carta Porte must be generated automatically from the ERP — not manually

Manual CFDI de traslado generation is a source of errors (incorrect RFC, incorrect merchandise description, outdated transporter permit number) and delays in shipping execution. Billing system integration with the WMS/TMS generates the CFDI automatically with correct data for each shipment.

WMS-TMS integration for carrier label generation eliminates the biggest shipping bottleneck

When the WMS automatically generates the transport order and the TMS automatically generates the carrier guide and label, shipping execution time is reduced from minutes to seconds per shipment.

Verify truck loading by scanner before closing the dock

Scanner verification that all documented packages are loaded in the truck (and no undocumented extra packages) is the final quality control of shipping execution. Without this verification, “forgotten” packages are only discovered when the customer reports not receiving their order.

The Carta Porte complement requires the transporter’s RFC and valid SICT permit

The Carta Porte complement requires specific transporter data (RFC, SICT permit number) that must be current in the carrier catalog. An expired permit number generates an invalid CFDI.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

> **Illustrative case: Shipping execution automation — DC with 840 shipments/day** — Current shipping process is 80% manual: guide generated in carrier portal, CFDI in the tax system, packing list printed from ERP.

Process step	Before (manual)	After (automated with TMS + tax integration)
Carrier guide and label generation	12 min/shipment · 2 dedicated FTEs	28 sec/shipment · Automatic generation from integrated WMS/TMS
CFDI de traslado + Carta Porte generation	8 min/shipment · RFC errors: 3.2% of CFDIs	42 sec/shipment · Data from ERP -> 0 RFC errors
Truck loading with scanner verification	Manual count — 18 min per truck of 80 packages	Scanner verification: 6 min per truck + 0 “forgotten” packages
Total shipping execution time per shipment	28 min + errors	5 min + 0 documentation errors

Result: Shipping labor savings: 2 FTEs. Shipping time per shipment: 28 min -> 5 min. SAT transport stops for missing Carta Porte: 0 in the 12 months post-implementation vs. 3 incidents in the prior 12 months.

HOW IT IS MEASURED

Shipping Documentation Accuracy % (shipments with complete and correct documentation)

Shipping Documentation Accuracy %(Shipments dispatched with correct guide + CFDI de traslado + Carta Porte + remittance / Total shipments) × 100

Benchmark: 100% documentation accuracy — any shipment without the correct CFDI is a fiscal risk A shipment without the CFDI de traslado or without the correct Carta Porte complement can be stopped by the SAT in an en-route inspection — with consequences of transport stoppage, fines, and reputational damage.

Shipping Execution Time (time from order ready-to-ship to truck departure)

Shipping Execution TimeAverage time from the last package of a truck being packed to the truck departing from the dispatch dock

Benchmark: <45 minutes for a standard operation · <15 minutes with fully automated shipping execution Slow shipping execution generates trucks departing late from the carrier pickup window — causing delivery delays and potential OTIF non-compliances that are not the carrier’s fault.

WHAT YOU WOULD USE

Shipping & CFDI

SAP EWM + SAP S/4HANA — Shipping Integration — Module: Automated Shipping Execution + CFDI de TrasladoSAP EWM + SAP S/4HANA integration automatically generates the remittance, CFDI de traslado with Carta Porte complement, and TMS communication for carrier guide generation.

CONTPAQi / Aspel (Siigo) — Module: CFDI de Traslado + Carta Porte GenerationMost widely used electronic billing systems in Mexico with support for CFDI de traslado generation with the current Carta Porte complement (version 3.1).

Carrier Labels

Shipstation / EasyPost — Module: Multi-Carrier Label Generation APIShipping label generation APIs that connect with multiple carriers and generate labels in the correct format for each carrier.

4 SO WHAT

Generate documents from data, verify loads by scan. Integrate WMS, TMS and billing, keep carrier data valid and hold documentation accuracy to 100%.

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